

Gestion d'une bibliothèque numérique

Objectif :

- Créer un index pour gérer une collection de livres.
- Indexer des documents représentant des livres avec différentes métadonnées.
- Répondre à des questions avec des requêtes Elasticsearch couvrant les notions du module.

Consignes de livrable

Pour chaque question fournir :

- La requête complète
 - Le résultat JSON collé
-

0. Identification du cluster (obligatoire, non notée)

Avant de commencer, exécuter cette commande et coller le résultat :

GET

```
/_cluster/stats?filter_path=cluster_name,cluster_uuid,nodes.count,indices.count,indices.docs
```

1. Création du Mapping pour l'index **library**

PUT /library

```
{
  "mappings": {
    "properties": {
      "title": {
        "type": "text",
        "analyzer": "standard"
      },
      "author": {
        "type": "keyword"
      },
      "publication_year": {
        "type": "date",
        "format": "yyyy"
      },
      "genre": {
        "type": "keyword"
      }
    }
  }
}
```

```

    "rating": {
      "type": "float"
    },
    "availability": {
      "type": "keyword"
    },
    "@timestamp": {
      "type": "date"
    }
  }
}

```

2. Indexation des livres via Bulk API

Voici un jeu de 20 livres à indexer dans Elasticsearch. Utilise la requête **POST** `/library/_bulk` pour indexer ces documents.

```

POST /library/_bulk
{"index": {"_index": "library", "_id": "1"}}
{"title": "Dune", "author": "Frank Herbert", "publication_year":
"1965", "genre": "Science Fiction", "rating": 4.8, "availability":
"Available", "@timestamp": "2024-09-30T10:00:00Z"}
{"index": {"_index": "library", "_id": "2"}}
{"title": "1984", "author": "George Orwell", "publication_year":
"1949", "genre": "Dystopian", "rating": 4.6, "availability":
"Checked Out", "@timestamp": "2024-09-30T11:00:00Z"}
{"index": {"_index": "library", "_id": "3"}}
{"title": "The Hobbit", "author": "J.R.R. Tolkien",
"publication_year": "1937", "genre": "Fantasy", "rating": 4.7,
"availability": "Available", "@timestamp": "2024-09-30T12:00:00Z"}
{"index": {"_index": "library", "_id": "4"}}
{"title": "To Kill a Mockingbird", "author": "Harper Lee",
"publication_year": "1960", "genre": "Fiction", "rating": 4.9,
"availability": "Available", "@timestamp": "2024-09-30T13:00:00Z"}
{"index": {"_index": "library", "_id": "5"}}
{"title": "Brave New World", "author": "Aldous Huxley",
"publication_year": "1932", "genre": "Dystopian", "rating": 4.5,
"availability": "Checked Out", "@timestamp": "2024-09-30T14:00:00Z"}
{"index": {"_index": "library", "_id": "6"}}

```

```

{"title": "The Great Gatsby", "author": "F. Scott Fitzgerald",
"publication_year": "1925", "genre": "Fiction", "rating": 4.4,
"availability": "Available", "@timestamp": "2024-09-30T15:00:00Z"}
{"index": {"_index": "library", "_id": "7"}}
{"title": "Moby Dick", "author": "Herman Melville",
"publication_year": "1851", "genre": "Fiction", "rating": 4.1,
"availability": "Available", "@timestamp": "2024-09-30T16:00:00Z"}
{"index": {"_index": "library", "_id": "8"}}
{"title": "Pride and Prejudice", "author": "Jane Austen",
"publication_year": "1813", "genre": "Romance", "rating": 4.9,
"availability": "Checked Out", "@timestamp": "2024-09-30T17:00:00Z"}
{"index": {"_index": "library", "_id": "9"}}
{"title": "The Catcher in the Rye", "author": "J.D. Salinger",
"publication_year": "1951", "genre": "Fiction", "rating": 4.3,
"availability": "Available", "@timestamp": "2024-09-30T18:00:00Z"}
{"index": {"_index": "library", "_id": "10"}}
{"title": "The Lord of the Rings", "author": "J.R.R. Tolkien",
"publication_year": "1954", "genre": "Fantasy", "rating": 4.9,
"availability": "Available", "@timestamp": "2024-09-30T19:00:00Z"}
{"index": {"_index": "library", "_id": "11"}}
{"title": "Harry Potter and the Philosopher's Stone", "author":
"J.K. Rowling", "publication_year": "1997", "genre": "Fantasy",
"rating": 4.7, "availability": "Checked Out", "@timestamp": "2024-
09-30T20:00:00Z"}
{"index": {"_index": "library", "_id": "12"}}
{"title": "The Shining", "author": "Stephen King",
"publication_year": "1977", "genre": "Horror", "rating": 4.6,
"availability": "Available", "@timestamp": "2024-09-30T21:00:00Z"}
{"index": {"_index": "library", "_id": "13"}}
{"title": "Frankenstein", "author": "Mary Shelley",
"publication_year": "1818", "genre": "Horror", "rating": 4.5,
"availability": "Available", "@timestamp": "2024-09-30T22:00:00Z"}
{"index": {"_index": "library", "_id": "14"}}
{"title": "Dracula", "author": "Bram Stoker", "publication_year":
"1897", "genre": "Horror", "rating": 4.4, "availability": "Checked
Out", "@timestamp": "2024-09-30T23:00:00Z"}
{"index": {"_index": "library", "_id": "15"}}
{"title": "War and Peace", "author": "Leo Tolstoy",
"publication_year": "1869", "genre": "Historical Fiction", "rating":
4.3, "availability": "Available", "@timestamp": "2024-10-
01T00:00:00Z"}

```

```
{"index": {"_index": "library", "_id": "16"}}
{"title": "The Odyssey", "author": "Homer", "publication_year": "-800", "genre": "Epic", "rating": 4.7, "availability": "Available", "@timestamp": "2024-10-01T01:00:00Z"}
{"index": {"_index": "library", "_id": "17"}}
{"title": "Crime and Punishment", "author": "Fyodor Dostoevsky", "publication_year": "1866", "genre": "Philosophical Fiction", "rating": 4.8, "availability": "Available", "@timestamp": "2024-10-01T02:00:00Z"}
{"index": {"_index": "library", "_id": "18"}}
{"title": "The Brothers Karamazov", "author": "Fyodor Dostoevsky", "publication_year": "1880", "genre": "Philosophical Fiction", "rating": 4.9, "availability": "Checked Out", "@timestamp": "2024-10-01T03:00:00Z"}
{"index": {"_index": "library", "_id": "19"}}
{"title": "Anna Karenina", "author": "Leo Tolstoy", "publication_year": "1877", "genre": "Romance", "rating": 4.7, "availability": "Available", "@timestamp": "2024-10-01T04:00:00Z"}
{"index": {"_index": "library", "_id": "20"}}
{"title": "Les Misérables", "author": "Victor Hugo", "publication_year": "1862", "genre": "Historical Fiction", "rating": 4.9, "availability": "Available", "@timestamp": "2024-10-01T05:00:00Z"}
```

3. Questions à résoudre par des requêtes Elasticsearch

A. Requêtes de base :

1. **Rechercher tous les livres disponibles dans la catégorie "Science Fiction" :**
 - Utilise une requête **term** sur le champ **genre** et **availability**.
2. **Chercher les livres publiés après 1950 :**
 - Utilise une requête **range** sur le champ **publication_year**.
3. **Rechercher un livre contenant le mot "New" dans son titre :**
 - Utilise une requête **match** sur le champ **title**.

B. Requêtes booléennes :

4. Trouver tous les livres dans le genre "Dystopian" avec une note supérieure à 4.5 : Combine une recherche booléenne avec **must** pour le genre et une requête **range** pour la note.
5. Trouver tous les livres sauf ceux de genre "Dystopian" : Utilise une requête booléenne avec **must_not**.

6. Trouver les livres qui sont disponibles ou ont une note supérieure à 4.7 : Utilise une requête booléenne avec la clause should.

C. Enrichissement des données :

7. Créer un enrichissement pour ajouter un champ popularity_tag à chaque livre en fonction de sa note : Notes supérieures à 4.8 = "Best Seller", Notes entre 4.5 et 4.8 = "Popular", Notes inférieures à 4.5 = "Regular"

D. Ré-indexation des données :

8. Ré-indexer les livres vers un nouvel index library_v2 et ajoute un champ edition avec la valeur "Original" à chaque document.

E. Transformations (Pivots et Latest) :

9. Créer une transformation pivot pour calculer combien de livres sont disponibles par genre.
10. Créer une transformation latest pour obtenir le livre le plus récemment publié par chaque auteur.

F. Dashboard Kibana

Créer un dashboard **Bibliothèque** avec au minimum 3 panels :

- **Panel 1 — Nombre de livres par genre** : bar chart ou pie chart.
- **Panel 2 — Distribution des notes** : histogram.
- **Panel 3 — Livres disponibles vs empruntés** : metric ou donut.

Fournir une capture du dashboard complet avec votre nom.